

QArm Lab Procedure

Tool Manipulation

Setup

1. Launch Quanser Interactive Labs and load the QArm Workspace.
2. Launch MATLAB and browse to the working directory for Lab 6 – Tool Manipulation.

Differential Kinematics

1. The [Welding.slx](#) in this lab is incomplete, Use the [Welding.slx](#) model from the previous lab.
2. Prior to running the model, open each model's [Configuration Parameters](#) and verify that they are configured as follows
 - a. Solver type: Fixed-step
 - b. Solver: ode4 (Runge-Kutta)
 - c. Fixed-step size (fundamental sample time): 500Hz
3. Run [Welding.slx](#) using the green Play button  under the Simulation Tab of your model. Monitor the time taken to go through the 4 sides of the rectangular welding trajectory. Is the time provided the same? If so, is the rate of motion for the end-effector uniform?
4. Close [Welding.slx](#). Open [Jacobian.slx](#). This model asks you to develop the manipulator's Jacobian matrix which relates the manipulator's joint speeds to the task speeds of its end-effector.

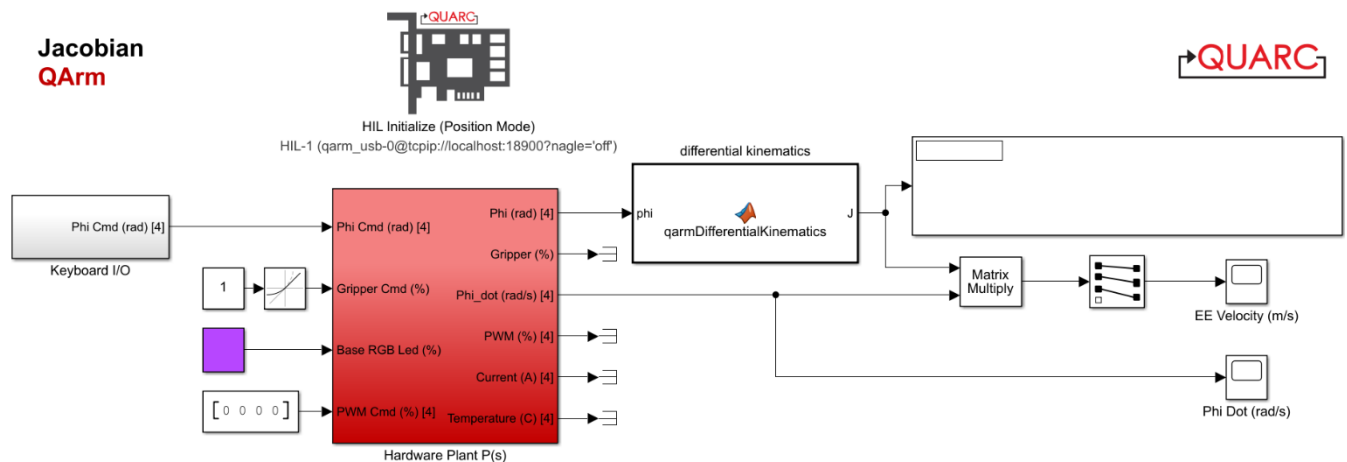


Figure 1.1 Jacobian Simulink Model

5. Double click on the embedded MATLAB function [qarmDifferentialKinematics](#). Lines 42 to 57 represent the 16 elements of the Jacobian matrix. Note that the Jacobian is provided to you in the [Tool Manipulation Application Guide](#). Complete this function and save it.
6. Run the model using the green Play button  under the Simulation Tab of your model and use the 4 arrow keys to control the x-y movement of the end-effector. Use Q and A keys to control the z direction movement of it. The display block shows the Jacobian matrix that your function calculates.

15. Vary the **max speed** variable in the Simulink model to speed up or slow down the algorithm. If you speed up the rate that the manipulator moves, you may have to relax the **threshold** for calculating whether you are close to the waypoint. Why is this?
16. Right click on the blank space in the Simulink model and select **Model Properties**. Click on **Callbacks** panel and select **InitFcn**. Change **waypoints** parameter to **waypoints_assembly**. Verify the rate and the trajectory is correct. You will see that the arm will perform like in an assembly line which will pick a part in a fixed position and assemble it to many places.
17. In your Simulink model, set the 'wait duration' constant to the amount of time the manipulator should wait at each location. The arm will now wait at each waypoint, which reflects the time needed to carry out any assembly related operations.

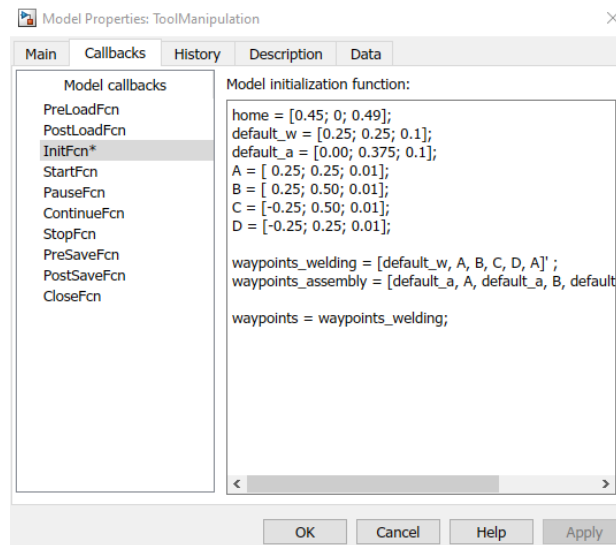


Figure 1.3 Model Properties Panel

18. Stop the model and close all experiments.